POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 6024A | Semester VI | Theory | 4 Credits

Computer Aided Design and Manufacturing

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Mechanical Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6024A as Computer Aided Design and Manufacturing in Semester VI for Mechanical Engineering. The official SITTTR detailed course page was unavailable. With the user's approved public-source approach, this handbook is transparently based on the NPTEL listing for IIT Delhi's Computer Aided Design and Manufacturing I and the detailed public Swayam CAD/CAM course preview. It does not claim to reproduce official REV2026 detailed-syllabus wording.

> **Source treatment:** Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. CAD/CAM files, CNC control, robotic/AGV systems and production databases must be used only under authorised account access, documented version control, verified programs and applicable machine-safety procedures.

> **Verified source note:** Verified sources support CAD purpose, 2D/3D drafting and modelling, curves/solids/mechanisms/assemblies, hardware/software and display/input/output; CAM, NC/CNC coordinate systems and motion control; FMS, AGVS, inventory/MIS/DSS and robotics. The four-module layout is a study sequence, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Computer Aided Design and Manufacturing connects digital product definition with computer-supported planning and machine control. It develops the ability to describe geometry, evaluate modelling choices, prepare manufacturing data and understand CNC/FMS/automation integration while preserving the verification and safety controls needed in real production.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□.

Prerequisite foundation

Engineering graphics, manufacturing processes, basic computer use, coordinate geometry, measurement awareness and safe workshop practice.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain CAD concepts, hardware/software, graphics and data representation.
2. Explain 2D/3D geometric, wireframe, surface and solid modelling for parts and assemblies.
3. Explain CAM concepts, NC/CNC coordinate systems, motion control and verified program preparation.
4. Explain flexible manufacturing, AGVS, inventory/data systems and robotics within integrated manufacturing.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO [illegible] exam-[illegible] [illegible] [illegible] [illegible] [illegible] [illegible]. Define, explain, apply [illegible] action words [illegible].

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain CAD purpose, graphics hardware/software, input/output and product-data concepts.	Understand
CO2	Create and interpret conceptual 2D/3D, wireframe, surface and solid-model representations.	Apply
CO3	Explain CAM, NC/CNC coordinate/motion concepts and the verified CAD-to-machine workflow.	Understand
CO4	Explain FMS, AGVS, inventory/information systems and robotics as integrated manufacturing elements.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7-PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	CAD Foundations, Graphics and Digital Product Data	CAD purpose and benefits; hardware/software; display, input and output devices; coordinate systems; data representation, modelling workflow and revision discipline.	Approx. 14 hours	CO1	Module 1
2	2D/3D Geometric, Wireframe, Surface and Solid Modelling	2D drafting; curves and primitives; transformations; wireframe, surface and solid models; features, assemblies and mechanisms; model comparison and application fit.	Approx. 16 hours	CO2	Module 2
3	CAM, NC/CNC and CAD-to-Machine Workflow	CAM purpose; process/toolpath data; NC/CNC principles; machine coordinate/reference systems; motion control; program verification, setup and controlled release.	Approx. 16 hours	CO3	Module 3
4	Flexible and Automated Manufacturing Integration	FMS components/types/layouts; AGVS concept and controls; computer inventory/MIS/DSS; robot types/anatomy/control/application; data, safety and performance perspective.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

CAD Foundations, Graphics and Digital Product Data

CAD purpose and benefits; hardware/software; display, input and output devices; coordinate systems; data representation, modelling workflow and revision discipline.

CO1 · Approx. 14 hours

2D/3D Geometric, Wireframe, Surface and Solid Modelling

2D drafting; curves and primitives; transformations; wireframe, surface and solid models; features, assemblies and mechanisms; model comparison and application fit.

CO2 · Approx. 16 hours

CAM, NC/CNC and CAD-to-Machine Workflow

CAM purpose; process/toolpath data; NC/CNC principles; machine coordinate/reference systems; motion control; program verification, setup and controlled release.

CO3 · Approx. 16 hours

Flexible and Automated Manufacturing Integration

FMS components/types/layouts; AGVS concept and controls; computer inventory/MIS/DSS; robot types/anatomy/control/application; data, safety and performance perspective.

CO4 · Approx. 14 hours

MODULE 1

CAD Foundations, Graphics and Digital Product Data

CAD purpose and benefits; hardware/software; display, input and output devices; coordinate systems; data representation, modelling workflow and revision discipline.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

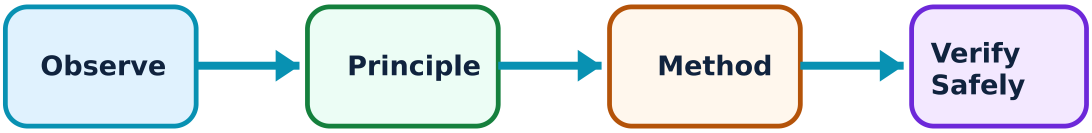
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for CAD Foundations, Graphics and Digital Product Data: identify the system, state its principle, follow the correct method and verify the result safely.

1. CAD role and workflow−

CAD uses computer systems to create, modify, analyse and optimise product representations. It supports clearer communication, design iteration, documentation and downstream integration when geometry, dimensions, material/metadata and revisions are controlled.

Study and application method

Map a design task from requirement to drawing/model release, identifying inputs, modelling decisions, review gates, output formats, revision identifier and downstream users.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Map a design task from requirement to drawing/model release, identifying inputs, modelling decisions, review gates, output formats, revision identifier and downstream users.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A visually correct model can still have missing tolerances, ambiguous references or the wrong revision. Use documented checking and release processes.

2. Graphics hardware, software and interfaces−

CAD environments combine hardware, operating systems, modelling applications, display/input devices and file/data interfaces. Device choice and data exchange influence user efficiency, precision, compatibility and traceability.

Study and application method

For a modelling task, identify appropriate input/output devices, software capability, data format, interoperability concern and an acceptable check before data is shared.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

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Common mistake / precaution: Do not install unapproved software, plug-ins or cloud synchronisation for production data. Protect licences, intellectual property and controlled files.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

2D/3D Geometric, Wireframe, Surface and Solid Modelling

2D drafting; curves and primitives; transformations; wireframe, surface and solid models; features, assemblies and mechanisms; model comparison and application fit.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

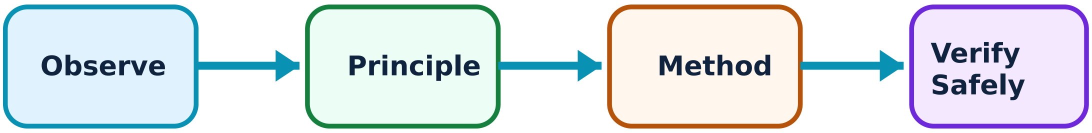
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for 2D/3D Geometric, Wireframe, Surface and Solid Modelling: identify the system, state its principle, follow the correct method and verify the result safely.

1. Geometry and model representations−

2D drawings communicate orthographic views and dimensions. 3D models may use wireframe, surface or solid representations: wireframes show edges, surfaces define skins and solids represent closed volume. Each suits different design, visualisation and manufacturing needs.

Study and application method

For an example component, identify which representation is appropriate at each stage and list the geometry/feature information that must be checked before using it downstream.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: For an example component, identify which representation is appropriate at each stage and list the geometry/feature information that must be checked before using it downstream.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Wireframe or surface data can look plausible while being open, self-intersecting or unsuitable for manufacture. Confirm topology/geometry with approved software checks.

2. Features, assemblies and motion awareness−

Feature-based solid modelling encodes design intent through sketches, constraints, extrusions, cuts, holes and patterns. Assemblies organise component relationships, while mechanism representations can help examine intended motion and interference conceptually.

Study and application method

Build or annotate a simple feature tree, identify design references/constraints, then perform a conceptual interference and assembly-sequence review.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Build or annotate a simple feature tree, identify design references/constraints, then perform a conceptual interference and assembly-sequence review.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: [illegible] concept [illegible] [illegible] [illegible]. [illegible] diagram / circuit / formula / procedure [illegible] [illegible]. [illegible] result [illegible] application [illegible] [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Common mistake / precaution: Digital interference checks do not replace physical validation for tolerance variation, deflection, assembly force, fastener behaviour or safety.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

CAM, NC/CNC and CAD-to-Machine Workflow

CAM purpose; process/toolpath data; NC/CNC principles; machine coordinate/reference systems; motion control; program verification, setup and controlled release.

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

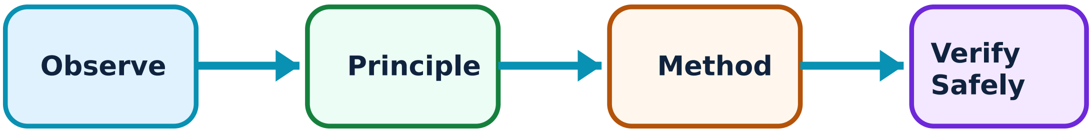
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for CAM, NC/CNC and CAD-to-Machine Workflow: identify the system, state its principle, follow the correct method and verify the result safely.

1. CAM and manufacturing data preparation−

CAM uses validated geometry, material, tooling, fixtures and process choices to assist in preparing operations and tool paths. It bridges design and machine execution but requires careful review of stock, datum, cutter access, collision risk, machining strategy and inspection needs.

Study and application method

Trace a conceptual CAM workflow: model import/check, stock/datum definition, tool/operation selection, path generation, simulation, post-processing, review and approved release.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Trace a conceptual CAM workflow: model import/check, stock/datum definition, tool/operation selection, path generation, simulation, post-processing, review and approved release.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Toolpath output is not automatically safe or correct. Controller compatibility, machine limits, tools, fixtures, offsets and setup must be verified before execution.

2. NC/CNC coordinate systems and motion−

NC/CNC systems execute programmed motion using defined axes, coordinate systems, interpolation and auxiliary functions. Accurate work/tool referencing and controlled motion are essential for dimensional outcome and collision avoidance.

Study and application method

Interpret an instructor-provided non-live coordinate example: identify machine/work coordinates, intended path, reference/datum assumptions and evidence needed before a supervised dry run.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Interpret an instructor-provided non-live coordinate example: identify machine/work coordinates, intended path, reference/datum assumptions and evidence needed before a supervised dry run.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Never run or edit code on production equipment without authorisation. Check program, work offset, tool data, clamp condition, guarding and prove-out procedure.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Flexible and Automated Manufacturing Integration

FMS components/types/layouts; AGVS concept and controls; computer inventory/MIS/DSS; robot types/anatomy/control/application; data, safety and performance perspective.

Approx. 14 hours **CO4** **Understand · Apply**

Learning goals

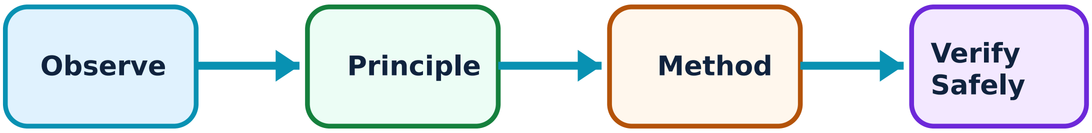
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Flexible and Automated Manufacturing Integration: identify the system, state its principle, follow the correct method and verify the result safely.

1. FMS and automated guided vehicle systems−

Flexible manufacturing systems coordinate programmable workstations, tools, handling and control to manage a family of parts. AGVs/AMRs can move materials using defined guidance, routing, charging and traffic-control strategies, subject to layout and safety constraints.

Study and application method

Map a notional FMS/AGV layout including stations, buffers, material flow, identification, control decisions, charging/maintenance point and protected human interfaces.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Map a notional FMS/AGV layout including stations, buffers, material flow, identification, control decisions, charging/maintenance point and protected human interfaces.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Mobile and automated equipment require engineered exclusion zones, traffic control, sensing, emergency response and authorised commissioning—never infer live settings from a study model.

2. Information systems and robotics−

Inventory systems, management information systems and decision-support systems organise manufacturing information for planning, control and improvement. Robots combine mechanisms, actuators, sensors and control for handling or process tasks; their usefulness depends on integration and safe deployment.

Study and application method

For a production case, identify the transaction/data needed at receipt, storage, issue, processing, inspection and dispatch; then state how a robot or automation aid would be evaluated.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: For a production case, identify the transaction/data needed at receipt, storage, issue, processing, inspection and dispatch; then state how a robot or automation aid would be evaluated.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Data access and robot programming are controlled activities. Do not make unattended changes to stock records, controller logic or robot paths without authorised review and safety validation.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: CAD Foundations, Graphics and Digital Product Data

Use the concept flow to revise observation, principle, method and verification.

Module 2: 2D/3D Geometric, Wireframe, Surface and Solid Modelling

Use the concept flow to revise observation, principle, method and verification.

Module 3: CAM, NC/CNC and CAD-to-Machine Workflow

Use the concept flow to revise observation, principle, method and verification.

Module 4: Flexible and Automated Manufacturing Integration

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Prepare a controlled CAD workflow map from requirement through revision/release to downstream CAM.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare wireframe, surface and solid representations for three example components.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Annotate a CAD-to-CAM process plan with geometry checks, datum, stock, tools, simulation and inspection points.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Interpret a supplied CNC coordinate/path example without uploading or executing code.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Design a conceptual FMS/AGV information-and-material-flow map that includes safety and human-interaction boundaries.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — CAD Foundations, Graphics and Digital Product Data

Explain CAD role and workflow and state one correct method, application or precaution.−

CAD uses computer systems to create, modify, analyse and optimise product representations. It supports clearer communication, design iteration, documentation and downstream integration when geometry, dimensions, material/metadata and revisions are controlled.

Answer framework: Map a design task from requirement to drawing/model release, identifying inputs, modelling decisions, review gates, output formats, revision identifier and downstream users.

Important point: A visually correct model can still have missing tolerances, ambiguous references or the wrong revision. Use documented checking and release processes.

Explain Graphics hardware, software and interfaces and state one correct method, application or precaution.−

CAD environments combine hardware, operating systems, modelling applications, display/input devices and file/data interfaces. Device choice and data exchange influence user efficiency, precision, compatibility and traceability.

Answer framework: For a modelling task, identify appropriate input/output devices, software capability, data format, interoperability concern and an acceptable check before data is shared.

Important point: Do not install unapproved software, plug-ins or cloud synchronisation for production data. Protect licences, intellectual property and controlled files.

Module 2 — 2D/3D Geometric, Wireframe, Surface and Solid Modelling

Explain Geometry and model representations and state one correct method, application or precaution.−

2D drawings communicate orthographic views and dimensions. 3D models may use wireframe, surface or solid representations: wireframes show edges, surfaces define skins and solids represent closed volume. Each suits different design, visualisation and manufacturing needs.

Answer framework: For an example component, identify which representation is appropriate at each stage and list the geometry/feature information that must be checked before using it downstream.

Important point: Wireframe or surface data can look plausible while being open, self-intersecting or unsuitable for manufacture. Confirm topology/geometry with approved software checks.

Explain Features, assemblies and motion awareness and state one correct method, application or precaution.−

Feature-based solid modelling encodes design intent through sketches, constraints, extrusions, cuts, holes and patterns. Assemblies organise component relationships, while mechanism representations can help examine intended motion and interference conceptually.

Answer framework: Build or annotate a simple feature tree, identify design references/constraints, then perform a conceptual interference and assembly-sequence review.

Important point: Digital interference checks do not replace physical validation for tolerance variation, deflection, assembly force, fastener behaviour or safety.

Module 3 — CAM, NC/CNC and CAD-to-Machine Workflow

Explain CAM and manufacturing data preparation and state one correct method, application or precaution.−

CAM uses validated geometry, material, tooling, fixtures and process choices to assist in preparing operations and tool paths. It bridges design and machine execution but requires careful review of stock, datum, cutter access, collision risk, machining strategy and inspection needs.

Answer framework: Trace a conceptual CAM workflow: model import/check, stock/datum definition, tool/operation selection, path generation, simulation, post-processing, review and approved release.

Important point: Toolpath output is not automatically safe or correct. Controller compatibility, machine limits, tools, fixtures, offsets and setup must be verified before execution.

Explain NC/CNC coordinate systems and motion and state one correct method, application or precaution.−

NC/CNC systems execute programmed motion using defined axes, coordinate systems, interpolation and auxiliary functions. Accurate work/tool referencing and controlled motion are essential for dimensional outcome and collision avoidance.

Answer framework: Interpret an instructor-provided non-live coordinate example: identify machine/work coordinates, intended path, reference/datum assumptions and evidence needed before a supervised dry run.

Important point: Never run or edit code on production equipment without authorisation. Check program, work offset, tool data, clamp condition, guarding and prove-out procedure.

Module 4 — Flexible and Automated Manufacturing Integration

Explain FMS and automated guided vehicle systems and state one correct method, application or precaution.−

Flexible manufacturing systems coordinate programmable workstations, tools, handling and control to manage a family of parts. AGVs/AMRs can move materials using defined guidance, routing, charging and traffic-control strategies, subject to layout and safety constraints.

Answer framework: Map a notional FMS/AGV layout including stations, buffers, material flow, identification, control decisions, charging/maintenance point and protected human interfaces.

Important point: Mobile and automated equipment require engineered exclusion zones, traffic control, sensing, emergency response and authorised commissioning—never infer live settings from a study model.

Explain Information systems and robotics and state one correct method, application or precaution.−

Inventory systems, management information systems and decision-support systems organise manufacturing information for planning, control and improvement. Robots combine mechanisms, actuators, sensors and control for handling or process tasks; their usefulness depends on integration and safe deployment.

Answer framework: For a production case, identify the transaction/data needed at receipt, storage, issue, processing, inspection and dispatch; then state how a robot or automation aid would be evaluated.

Important point: Data access and robot programming are controlled activities. Do not make unattended changes to stock records, controller logic or robot paths without authorised review and safety validation.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: CAD Foundations, Graphics and Digital Product Data.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.

3. State one key definition from Module 2: 2D/3D Geometric, Wireframe, Surface and Solid Modelling.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: CAM, NC/CNC and CAD-to-Machine Workflow.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Flexible and Automated Manufacturing Integration.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: CAD purpose and benefits; hardware/software; display, input and output devices; coordinate systems; data representation, modelling workflow and revision discipline..
2. Develop a complete answer or practical plan for: 2D drafting; curves and primitives; transformations; wireframe, surface and solid models; features, assemblies and mechanisms; model comparison and application fit..
3. Develop a complete answer or practical plan for: CAM purpose; process/toolpath data; NC/CNC principles; machine coordinate/reference systems; motion control; program verification, setup and controlled release..
4. Develop a complete answer or practical plan for: FMS components/types/layouts; AGVS concept and controls; computer inventory/MIS/DSS; robot types/anatomy/control/application; data, safety and performance perspective..

LAST-MILE REVISION

Quick Revision

Module 1: CAD Foundations, Graphics and Digital Product Data

CAD purpose and benefits; hardware/software; display, input and output devices; coordinate systems; data representation, modelling workflow and revision discipline.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: 2D/3D Geometric, Wireframe, Surface and Solid Modelling

2D drafting; curves and primitives; transformations; wireframe, surface and solid models; features, assemblies and mechanisms; model comparison and application fit.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: CAM, NC/CNC and CAD-to-Machine Workflow

CAM purpose; process/toolpath data; NC/CNC principles; machine coordinate/reference systems; motion control; program verification, setup and controlled release.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Flexible and Automated Manufacturing Integration

FMS components/types/layouts; AGVS concept and controls; computer inventory/MIS/DSS; robot types/anatomy/control/application; data, safety and performance perspective.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
CAD role and workflow	CAD uses computer systems to create, modify, analyse and optimise product representations.
Graphics hardware, software and interfaces	CAD environments combine hardware, operating systems, modelling applications, display/input devices and file/data interfaces.
Geometry and model representations	2D drawings communicate orthographic views and dimensions.
Features, assemblies and motion awareness	Feature-based solid modelling encodes design intent through sketches, constraints, extrusions, cuts, holes and patterns.
CAM and manufacturing data preparation	CAM uses validated geometry, material, tooling, fixtures and process choices to assist in preparing operations and tool paths.
NC/CNC coordinate systems and motion	NC/CNC systems execute programmed motion using defined axes, coordinate systems, interpolation and auxiliary functions.
FMS and automated guided vehicle systems	Flexible manufacturing systems coordinate programmable workstations, tools, handling and control to manage a family of parts.
Information systems and robotics	Inventory systems, management information systems and decision-support systems organise manufacturing information for planning, control and improvement.

LEARNING RESOURCES

References

Recommended CAD/CAM references

1. P. N. Rao, CAD/CAM: Principles and Applications.
2. I. Zeid, CAD/CAM Theory and Practice.
3. M. P. Groover and E. W. Zimmers, CAD/CAM: Computer-Aided Design and Manufacturing.
4. M. P. Groover, Automation, Production Systems, and Computer-Integrated Manufacturing.

Approved public CAD/CAM sources

- https://nptel.ac.in/courses/112102102
- https://onlinecourses.swayam2.ac.in/nou22_me08/preview

These sources provide the verified study basis while official Course 6024A detail is unavailable; they do not replace licensed software guidance, controller manuals, approved manufacturing data or authorised machine-operation procedures.